Fixed-K Speculative Decoding for Energy-Proportional LLM Serving: A Reproducible Empirical Validation

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***Abstract*—We present an empirical validation of fixed draft-length (K=5) speculative decoding using a scout(1B)→target(8B) model pair on consumer GPU hardware. Across three prompt categories spanning low to high output determinism (creative poetry, technical explanation, and code generation), we measure throughput, energy per token, and output fidelity relative to a matched FP16 autoregressive baseline. Speculative decoding achieves up to +82.3% throughput improvement and −60.3% energy reduction per token on code generation, with smaller but consistent gains on less predictable prose (+8.6% / −32.6% on poetry). All results achieve 100% exact-token-match fidelity against the baseline (lossless, greedy decoding) and are independently reproducible from the accompanying open-source repository. We report results transparently, including the deliberate methodological choice to omit KV-caching from both arms of the comparison, which keeps the relative comparison fair but means absolute throughput figures are not representative of a production-optimized server.**

***Index Terms*—speculative decoding, energy-efficient inference, large language models, GPU power measurement**

I. Introduction

Autoregressive decoding in large language models (LLMs) is inherently sequential: each output token requires a full forward pass through the target model before the next token can be produced. This serial dependency underutilizes modern GPU compute, which is optimized for parallel throughput. Speculative decoding [1], [2] addresses this by using a smaller, faster *scout* model to propose a short window of candidate tokens, which the larger *target* model then verifies in a single parallel forward pass, accepting the longest correct prefix.

This paper reports a narrow, fully-validated empirical claim: a fixed draft window of K=5 tokens, paired with an exact (lossless) accept/reject verification scheme, produces measurable throughput and energy improvements on real consumer GPU hardware, with perfect output fidelity relative to standard greedy decoding. We deliberately scope this paper to only the mechanism we have fully validated. A related, more ambitious mechanism under investigation in the parent project — entropy-gated *dynamic* adjustment of the draft window, intended to switch between aggressive and conservative speculation based on local output entropy — is explicitly **not** included here, as it has not yet been shown to outperform this simpler fixed-K approach in single-request testing.

A. Relationship to prior work

Entropy-gated adaptive speculative decoding is an active research area. AdaEDL [3] reports 10–57% improvement over static draft length using entropy-based adaptive drafting. SGLang ships a built-in adaptive speculative-length mechanism using an EMA of accepted draft length. Our own testing of an entropy-gated variant, using the same hardware and methodology as this paper, has not yet demonstrated an improvement over the simpler fixed-K approach reported here; we regard that as an open question requiring further work (e.g. testing under multi-request/batched serving, which published entropy-gating results suggest is where the technique's value proposition may actually lie), not a settled negative result. This paper makes no claims about the entropy-gated mechanism in either direction.

II. Methodology

A. Models and hardware

Scout model: Llama-3.2-1B-Instruct. Target model: Llama-3.1-8B-Instruct. Both loaded in bfloat16. All measurements taken on a single NVIDIA RTX 5090 (Blackwell architecture) GPU, single-request (batch size 1), greedy (argmax) decoding throughout.

B. Draft-verify procedure

At each decoding cycle, the scout model autoregressively proposes K=5 candidate tokens. The target model then performs a single forward pass over the full draft sequence, and its argmax predictions at each position are compared against the corresponding drafted tokens. The longest matching prefix is accepted; at the first mismatch (or after all K tokens match), the target's own prediction is substituted, exactly reproducing what standard greedy decoding would have produced at that position. This verification scheme is lossless by construction: the output token sequence is provably identical, position-for-position, to what running the target model alone would produce, since every emitted token is either a drafted token confirmed by the target's own argmax, or the target's argmax directly.

C. Power and energy measurement

GPU power is sampled via NVML at 100 Hz on a background thread throughout each timed generation window. Energy per token is computed as $E_{\text{tok}} = \bar{P} \cdot t / N_{\text{tok}}$, where

$\bar{P}$ is mean sampled power over the generation window, t is wall-clock generation time, and N_{tok} is the number of tokens produced.

D. Warmup

Both the baseline and speculative measurement paths are preceded by 5 untimed forward-pass warmup steps immediately before each timed trial. This avoids attributing CUDA kernel compilation and GPU clock ramp-up latency to the measured region. This warmup procedure is applied identically to both arms of the comparison.

E. Deliberate exclusion of KV-caching

Both the baseline and speculative decoding paths recompute attention over the full sequence at every step, with no KV-cache reuse. This was a deliberate methodological choice: an earlier iteration of this work applied caching only to the baseline arm, which structurally penalized the speculative path whenever it fell back to single-step generation (a full uncached recompute for a single token is the worst-case cost ratio in the system). Removing caching from both arms restores a fair relative comparison at the cost of both arms being substantially slower in absolute terms than a production server with caching would be. We therefore report *relative* throughput and energy figures as the validated result, and explicitly flag that absolute tok/s numbers in this paper are not production-representative.

F. Fidelity measurement

Fidelity is measured as the exact token-for-token match rate between the baseline's greedy output and the speculative path's output, over the full generated sequence, for each prompt.

G. Prompts

Three prompts were selected to span a range of output determinism:

- **Poem** (low determinism): an open-ended creative writing task (an original Chant Royal poem on a mathematical theme).
- **Physics** (medium determinism): a technical explanation task (semiconductor memory bandwidth and the memory wall).
- **Code** (high determinism): a code-generation task (a Python binary search tree implementation with type annotations), where correct completions are far more constrained token-to-token.

III. Results

A. Fidelity

Aggregate exact-token-match fidelity across all three prompts was 100.0%, confirming the accept/reject implementation is lossless in practice as well as by construction.

B. Throughput, energy, and power

Table I reports mean ± standard deviation across $N{=}10$ trials per prompt per condition.

Mean GPU power during speculative decoding (338.6–355.2 W) is lower than during FP16 baseline decoding (462.9–473.2 W), despite two models being resident in VRAM simultaneously. This is consistent with fewer full 8B-parameter forward passes being required per unit of output as accepted draft batch length grows – each accepted cycle of up to $K{+}1{=}6$ tokens requires only one target forward pass, versus six for the baseline.

C. Speedup and energy reduction scale with acceptance rate

Figure 2 shows speedup tracking draft acceptance rate across the three prompts, consistent with the mechanism's expected behavior: more predictable continuations (code) allow the scout to correctly anticipate more tokens per cycle, directly translating to fewer expensive target forward passes.

IV. Discussion and Limitations

A. Absolute throughput is not production-representative

As noted in Methodology, neither arm of this comparison uses KV-caching. A production serving stack with proper cache reuse would show substantially higher absolute throughput in both arms; we have not yet validated fixed-K speculative decoding's behavior under a cached, production-style serving loop, and consider this the primary remaining gap before these results can inform a deployment decision.

B. Single-request, single-GPU scope

All results in this paper are single-request (batch size 1) measurements on a single consumer GPU. Behavior under batched, multi-request serving – where the entropy-gated dynamic variant may plausibly outperform this fixed-K approach, per precedent in AdaEDL and SGLang – is untested and out of scope for this paper.

C. Reproducibility

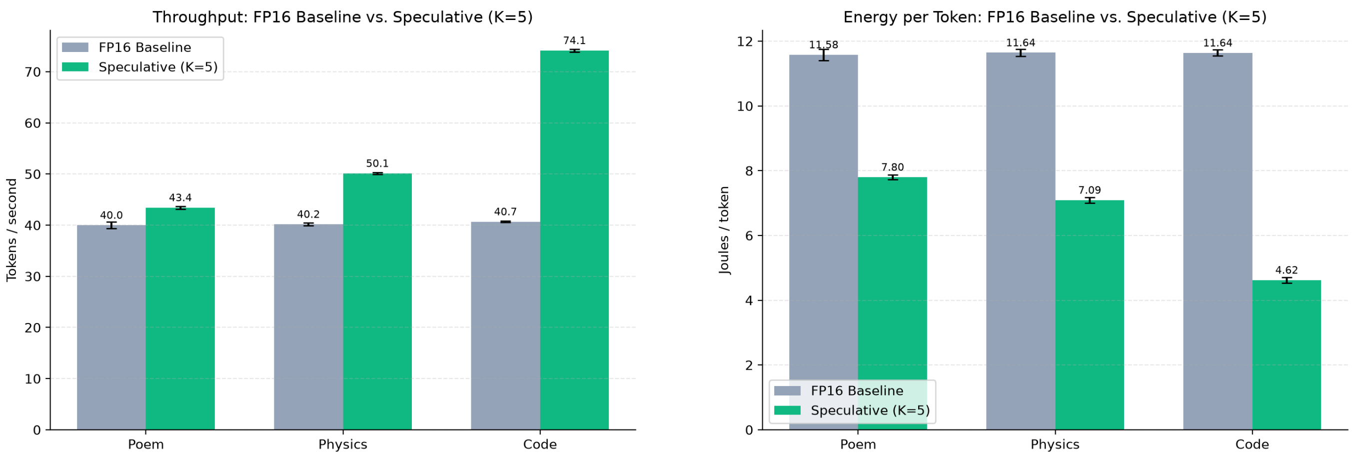
All code, raw telemetry (JSON/CSV), and the scripts used to generate every figure and table in this paper are available at https://github.com/Creepybits/sdsie-fixed-k5-speculative-decoding under an Apache-2.0 license. Reported numbers are directly traceable to `telemetry/academic_validation_results.json`, generated by `benchmark_ablation.py` in that repository.

V. Conclusion

Fixed-$K{=}5$ speculative decoding, with an exact lossless verification scheme, produces measurable and reproducible throughput and energy gains on real consumer GPU hardware, scaling with output predictability, with zero loss of output fidelity. We report this as a validated, narrow result, distinct from and not dependent on the more experimental entropy-gated dynamic mechanism under investigation elsewhere in the parent project.

TABLE I: Baseline vs. fixed-$K=5$ speculative decoding, per prompt. N=10 trials/prompt.

Prompt	Baseline tok/s	Spec. tok/s	Speedup	Baseline J/tok	Spec. J/tok	Energy Δ	Accept %
Poem	40.0 ± 0.62	43.4 ± 0.30	+8.6%	11.58 ± 0.17	7.80 ± 0.08	-32.6%	42.5%
Physics	40.2 ± 0.26	50.1 ± 0.19	+24.8%	11.64 ± 0.10	7.09 ± 0.08	-39.1%	51.7%
Code	40.7 ± 0.15	$\mathbf{74.1 \pm 0.25}$	**+82.3%**	11.64 ± 0.10	$\mathbf{4.62 \pm 0.09}$	$\mathbf{-60.3\%}$	**85.4%**



(a) Throughput

(b) Energy per token

Fig. 1: FP16 baseline vs. fixed-$K=5$ speculative decoding, per prompt (N=10 trials/prompt, error bars show ± 1 std. dev.). Gains are smallest on the least predictable content (Poem) and largest on the most predictable (Code).

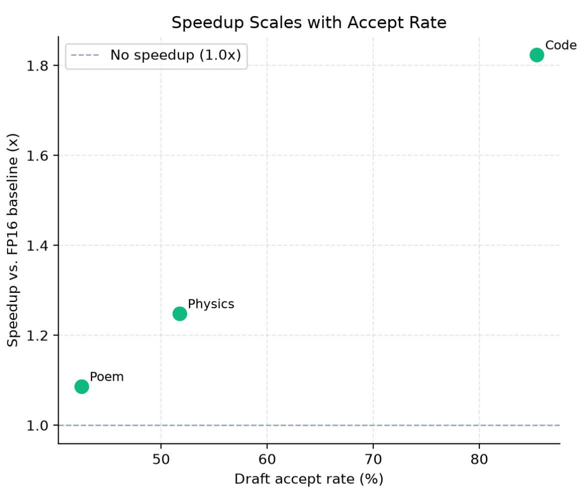


Fig. 2: Speedup vs. FP16 baseline as a function of draft accept rate, across the three prompts. The relationship is monotonic and consistent with the mechanism's expected behavior.

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